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Modeling and prediction of desalination performance in a scaled-up membrane capacitive deionization system using machine learning and deep learning

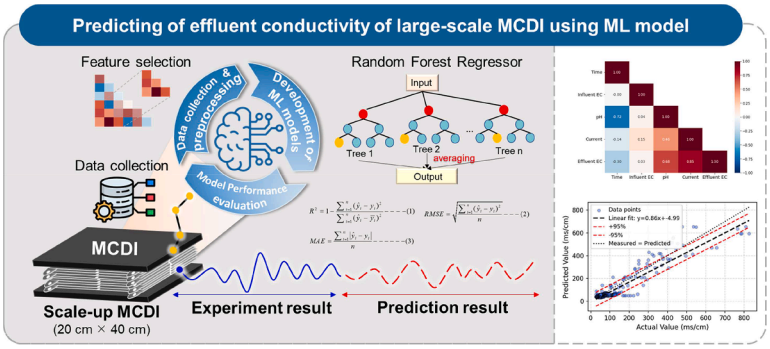
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HIGHLIGHTS

- A scale-up MCDI system was designed to operate varying influent conductivity.
- Machine learning model is employed to predict MCDI effluent conductivity.
- Hyperparameter with restricted grid search approach optimized model performance.
- Random Forest model surpassed deep learning model for MCDI data prediction.
- SHAP analysis revealed key feature contributions from "black box" of ML models.

GRAPHICAL ABSTRACT



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Keywords:
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ABSTRACT

Membrane capacitive deionization (MCDI) is a promising desalination technology characterized by low energy consumption. However, its practical application often relies heavily on trial-and- error approaches, which are time consuming and inefficient for operation tuning. Since traditional mechanistic models tend to oversimplify complex physicochemical interactions, data-driven modeling plays an essential role in predicting the performance of MCDI system. In this study, a scaled-up MCDI system was operated under varying influent condictions, which served as dataset for the machine learning (ML) and deep learning (DL) models, to establish a robust data-driven modeling farmework. Four ML and DL models, including random forest (RF), extreme gradient boost (XGBoost), multilayer perceptron (MLP), and long-short term memory (LSTM), were developed to predict effluent conductivity (EC_{out}). To ensure a leakage-free assessment, a strictly chronological cycle-based data splitting strategy was adopted. Additionally, hyperparameter tuning was conducted to balanced predictive performance and model stability, while Shapley Additive Explanations (SHAP) were employed to decode the underlying physicochemical logic of the black-box models. Among these models, the RF model achieved superior

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robustness and stable predictive accuracy, with a coefficient of determination (R^2) of 0.79, root mean square error (RMSE) of 36.25 μs/cm, and a mean absolute error (MAE) of 22.92 μs/cm. The SHAP analysis validated the model's interpretability by identifying time, current, and pH as the most influential input features, thereby revealed that current as the driving force for electromigration, and pH as a critical indicator of electrochemical stability. Overall, this study demonstrates the potential of ML/DL models to capture nonlinear dynamics in large-scale MCDI systems, providing a bridge between data-driven predictions and process understanding.

1. Introduction

With the rapid growth of industrialization, urbanization, and population, water scarcity has attracted increasing global attention in recent years (Wei et al., 2018). The water-energy nexus highlights the complex interconnections between energy and water sectors, and has become a key issue for achieving sustainable development (Gleick et al., 1994). In light of this global environmental backdrop, the development of desalination technologies, particularly those emphasizing low energy consumption, has emerged as a critical area of research. Compared to the conventional desalination technologies (e.g., thermal-driven and pressure-driven processes), membrane capacitive deionization (MCDI), an advanced form of capacitive deionization (CDI), has been regarded as a promising technique for the desalination of brackish water or low-salinity solution. Its advantages include low energy consumption, high water recovery, reduced fouling and scaling, operational flexibility, and environmental friendliness (Liu et al., 2024; Hsieh et al., 2024; Yoon et al., 2019; Chen et al., 2022).

With the continued development of MCDI technology, applications have progressed from laboratory studies to pilot-scale demonstrations. To date, scaled-up MCDI systems have been explored for hard water softening, wastewater reclamation and ammonia-nitrogen recovery, demonstrating practical feasibility across a range of water treatment scenarios (Chen et al., 2025; Yoon et al., 2024; Chen et al., 2022; Wang et al., 2022; Shen et al., 2021; Kim et al., 2019). For example. a pilot-scale MCDI unit with a treatment capacity of 21.1 m^3/day achieved stable ion removal efficiency exceeding 80 % with a low specific energy consumption of 0.21 ± 0.07 kWh/m^3 (Chen et al., 2025). Although MCDI has proven feasible and effective in real-site applications, its desalinization performance remains highly dependent on operating parameters. In practical, parameter adjustments are frequently guided by empirical knowledge and trial and error approach.

Traditional mechanistic models, such as modified Donnan (mD) model and the Nernst-Plank equation, have been widely employed to simulate ion transport and adsorption behaviors (Biesheuvel et al., 2009; Biesheuvel et al., 2011), and to support process optimization. However, these models often overlook nonlinear relationships among operational variables and only provide a limited representation of physicochemical interactions to reduce computational complexity (Saffarimiandoab et al., 2025). For example, theoretical frameworks like lattice Boltzmann simulations (Liu et al., 2022) or modified Donnan models (Zhao, 2013) typically simplify some parameters, such as pH fluctuations, as constants. Consequently, they may fail to capture the complex physicochemical dynamics inherent in real-world, scaled-up operations. In contrast to traditional mechanistic approaches, machine learning (ML) models have emerged as a promising alternative due to their strong capability to capture system complexity and nonlinear behaviors (Zhu et al., 2023; Huang et al., 2021). By learning directly from experimental data, ML models provide a flexible and scalable pathway for enhancing MCDI performance.

Recently, ML models have been applied to a wide range of MCDI-related tasks, including data prediction, electrode performance optimization, and process optimization (Kong et al., 2025; Chae et al., 2024; Son et al., 2023; Zhu et al., 2022; Saffarimiandoab et al., 2021). Nevertheless, most ML applications remain confined to lab-scale cell featuring an effective area of approximately 7 cm × 7 cm (e.g., Son et al., 2023; Chae et al., 2024), whereas only limited studies investigating pilot-scale systems (Zhu et al., 2022). Scaling up to commercial dimensions introduces heterogeneity in flow distribution and electrical fields that may be negligible at the lab scale but critical for industrial deployment. Zhu et al. (2022) investigated a pilot-scale MCDI system, in which Random Forest (RF) and Multilayer Perceptron (MLP) models were employed to predict cycle time based on multiple input variables under commercial MCDI operating conditions, and to optimize operational parameters for enhancing production efficiency. The authors suggested that future research should explore more advanced Deep Learning (DL) algorithms, such as long short-term memory (LSTM) to further improve predictive accuracy. In addition, many ML approaches are inherently black-box in nature, relying solely on input and output relationships without revealing the underlying logic behind their predictions (Pan et al., 2022). Although such black-box models can achieve high predictive accuracy, their lack of transparency limits their applicability in process control and decision-making (Mi et al., 2020). Striking a balance between interpretability and accuracy has therefore become increasingly important. Note that there is a growing need to develop comprehensive ML models that not only provide accurate predictions but also offer interpretability.

In this study, a scaled-up MCDI system operates under varying influent condictions, served as dataset, and further developed a detailed ML/DL modeling framework, comprising data preprocessing, model development, and model evaluation and interpretation, which we consider more comprehensive and appropriate for large-scale MCDI systems to achieve effective prediction. Furthermore, establishing a reliable ML workflow is a necessary first step before extending the framework toward long-term performance, derived indicators and cycle-level behavior. Feature variables were selected based on correlation analysis between the input and output data, and four ML/DL models were employed to predict the effluent conductivity (EC_{out}), namely Random Forest (RF), Extreme Gradient Boost (XGBoost), Multilayer Perceptron (MLP), and Long Short-Term Memory (LSTM). The predictive accuracy of each model was evaluated using three metrics: coefficient of determination (R^2), the root mean squared error (RMSE), and the mean absolute error (MAE). Subsequently, the hyperparameter tuning was conducted to balance predictive performance and model stability. Furthermore, to bridge the gap between data-driven predictions and process understanding, the contribution of input features was interpreted using SHAP analysis. This step aimed to decode the underlying physicochemical logic learned by the model, thereby demonstrating its alignment with fundamental MCDI principles. Overall, this study implemented ML/DL-based frameworks that enabled accurate and interpretable prediction of EC_{out}, supporting to extend the framework toward long-term performance and parameter optimization of scaled-up MCDI systems.

2. Materials and methods

2.1. The scaled-up MCDI stack

2.1.1. MCDI stack design and set-up

In the MCDI module, each electrode pair had an effective area of 20 cm × 40 cm and was fabricated by coating activated carbon powder (ACS 20, China Steel Chemical Co., Taiwan) onto titanium current collectors. A 2 mm woven polyethylene mesh sheet was placed between the electrodes to ensure uniform distribution of the feed solution. The MCDI

module also incorporated an anion exchange membrane (AEM, A11, Beijing Jie Rui Environmental Science and Technology Co., China) adjacent to the anode, and a cation exchange membrane (CEM, C11, Beijing Jie Rui Environmental Science and Technology Corporation, China) alongside the cathode.

2.1.2. MCDI experiment process

In this present work, all experiments were conducted in batch mode, which is similar to that used in our previous work (Liu et al., 2019). The system consisted of a MCDI module with 6 electrode pairs, a peristaltic pump (SF-5248-400, Super Flow, Taiwan), a conductivity meter (Orion™ Versa Star Pro™ with an Orion 013010MD probe, Thermo Scientific, USA), a pH meter (Orion™ Versa Star Pro™ with an Orion 8157BNUMD probe, Thermo Scientific, USA), and a power supply (PSW 30–108, GW Instek, Taiwan), as depicted in Fig. 1 (a) and (b). During the charging step, a constant voltage of 2 V was applied for 350 seconds, followed by a discharging step at 0 V for 188 seconds. Feed solution with different NaCl concentrations was pumped into the MCDI stack with a total solution volume of 1 L. The variation in conductivity and pH were continuously monitored and recorded using a connected computer. Notably, to imporve the generalizability of the models and expend the operation domain, we conducted a series of experiments with varying influent concentration using scaled-up MCDI system. Fig. 1 (c) presents the influent and effluent conductivity for each cycle under different influent conditions. In addition, Fig. 1 (d) illustrates the temporal variation of effluent conductivity in a single charging phase. Details of the MCDI operating conditions are summarized in Table S1.

2.2. ML/DL modeling process

In this study, the effluent conductivity (EC_{out}) prediction process was established in flowing steps: data preprocessing, model development, and mode evaluation and interpretation (Fig. 2). Details of these steps are described below.

2.2.1. Data collection and preprocessing

Data were collected every 3 seconds across 47 independent experimental runs, with each run encompassing a single 350-second charging cycle. This consistency yielded approximately 116 time-steps per cycle, resulting in a total dataset of 5,452 data points. Among these experiment, five types of data, including Time (T), influent conductivity (EC_{in}), pH, current(A), and effluent conductivity (EC_{out}), were collected and recorded.

Prior to model development, the collected data were preprocessed to ensure a smooth and reliable modeling process. The preprocessing workflow included several standard steps: data cleaning, data

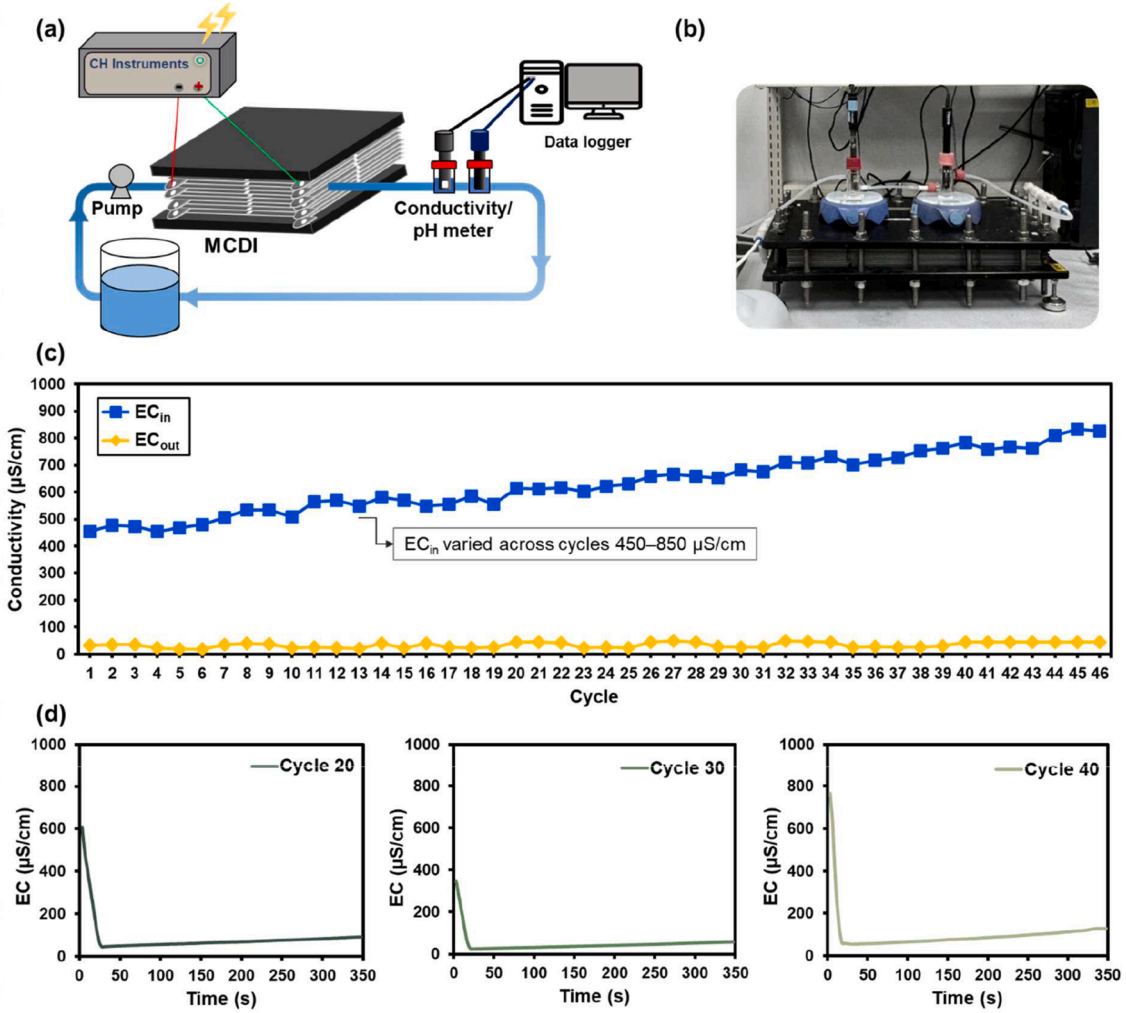


Fig. 1. MCDI Experiment for desalination performance. (a) and (b) Schematic illustration of the MCDI system used in this study. An influent stream is pumped through the MCDI module. A power supply provides a constant voltage to drive ion adsorption process. Effluent conductivity and pH are real-time monitored by sensors connected to a data logger. A photograph of the scale-up MCDI system also shown. (c) presents the influent and effluent conductivity for each cycle under different influent conditions. To broaden the operating range and enhance model generalizability, EC_{in} was varied between 450 and 850 μS/cm across cycles to cover multiple influent conditions. (d) profiles of conductivity performance within a single charging phase.

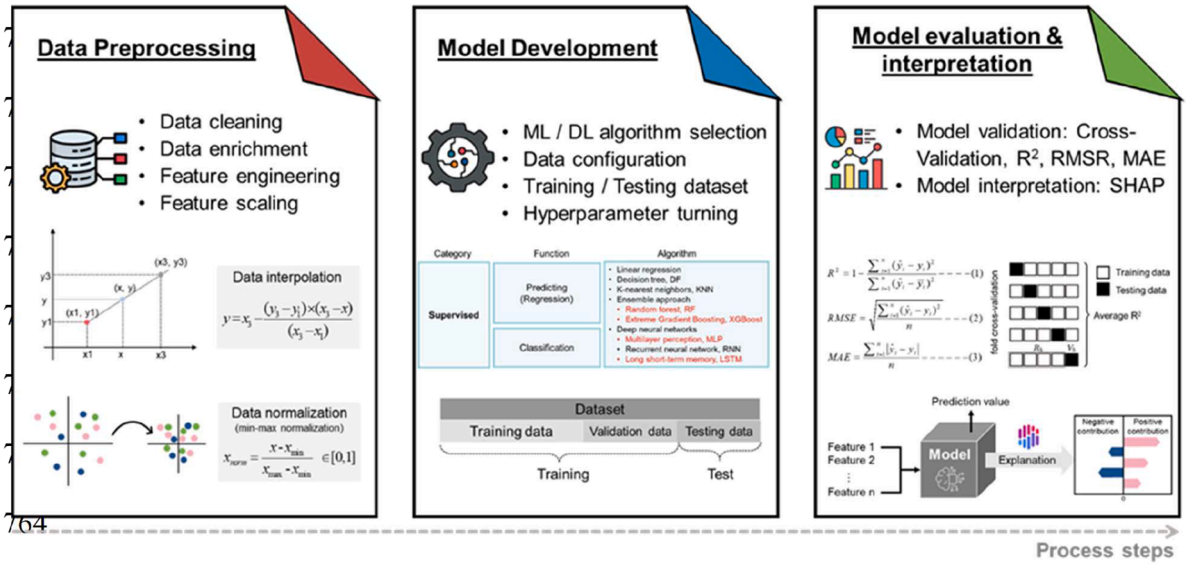


Fig. 2. Flowchart of the ML modeling for predicting effluent conductivity in the MCDI system. The framework comprises three stages: data preprocessing, model development, and mode evaluation and interpretation. In the preprocessing stage, raw experimental data underwent cleaning, enrichment, interpolation, and feature scaling to address missing values and ensure uniform input ranges. The model development stage involved selecting appropriate models, including RF, XGBoost, MLP, and LSTM, and configuring optimal hyperparameters through a restricted grid search. In the final stage, model performance was evaluated using R^2, RMSE, and MAE metrics. Additionally, SHAP was applied to interpret the models by quantifying the contribution of each input feature, thereby enhancing interpretability of the predictive framework.

enrichment, feature engineering, and feature scaling (Zhu et al., 2023). Data cleaning involved two main procedures as outlier removal and gap filling. Outliers usually arise from instrument malfunction or equipment faults. Once detected, these values were excluded from the dataset. Missing data points, constituting approximately 4.7 % of the total dataset in this study, were primarily attributed to the latency of the conductivity sensor relative to the rapid ion absorption process. During the MCDI experiments, data gaps occurred mainly during the initial 10–20 seconds of the charging phase, where rapid ion adsorption caused conductivity to drop faster than the response time of the EC sensor.

To address this, first-order Lagrange polynomial interpolation, which is equivalent to linear interpolation between adjacent time points, was applied to fill short gaps in the effluent conductivity time series. Besides, Abidin et al. (2018) claimed the approaches of imputing missing values, percentages of missing data points and sizes of datasets effect model performance. To ensure robustness, a high-order Lagrange polynomial was employed as a control group for comparison. Both approaches were conducted using the same dataset, characterized by the same missing points distribution gap locations. If the time-series data are presented as (x_1, y_1), …, (x_n, y_n), the linear interpolation formula is expressed as shown in Eq. (1), and the high-order Lagrange polynomial formula is shown in Eq. (2) (Walker et al., 2020).

$$y = y_j \frac{x - x_{j+1}}{x_i - x_{j+1}} + y_{j+1} \frac{x - x_j}{x_i - x_j} \tag{1}$$

$$y = \sum_{i=0}^{n} y_i \prod_{j=0, j\neq i}^{n} \frac{x - x_j}{x_i - x_j} \tag{2}$$

Feature selection was performed to identify the highly relevant input variables while minimizing the risk of overfitting. In this study, we adopted a hybrid strategy combining physical domain knowledge and statistical metrics (e.g., Pearson correlation coefficient, PCC). Initially, feature candidates were determined based on established MCDI theoretical mechanisms (Tang et al., 2017; Wang et al., 2019), and prior experimental experience (Hou et al., 2013; Liu et al., 2019; Chen et al., 2025), selecting retained features. Subsequently, PCC analysis was conducted to evaluate the linear relationships among variables by quantifying the strength and direction of their associations (Chou et al., 2024; Sun et al., 2023). The correlation coefficient (r), was calculated using Eq. (3).

$$r = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2 \sum_{i=1}^{n}(y_i - \overline{y})^2}} \tag{3}$$

where x_i (or y_i) represents individual data points of each feature, and$\overline{x}$ ($\overline{y}$) represents the corresponding mean value.

The value of r ranges from –1 to 1, with 1 indicating a strong positive correlation, –1 a strong negative correlation, and 0 indicating no correlation. In addition, the SFR was used to ensure that the number of training samples per feature was sufficient for reliable model development. According to Zhu et al. (2023), high SFR (≥100) were observed in time-series studies, and higher SFRs are generally required to support confident model training. Feature scaling was the final preprocessing step and was essential for good modeling practices (Xu et al., 2023). Since variable magnitudes may differ substantially, normalization ensures that all features contribute proportionally to the model (Saffarimiandoab et al., 2021). In this study, all data points were rescaled to a range between 0 and 1 using min-max normalization, as shown in Eq. (4).

$$x_{norm} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{4}$$

where the x, x_{min}, and x_{max} were the entry, minimum, and maximum values of the dataset for the corresponding input feature.

2.2.2. ML/DL modeling approaches

This study adopted a cycle-based chronological strategy to split the dataset into training and testing sets. Model overfitting may occur when a model captures noise or idiosyncrasies in the training data, resulting in degraded generalization to unseen data (Chang et al., 2022). In time-series datasets, an additional risk is information leakage when randomly shuffled splits allow temporally adjacent observations from the same operational cycle to appear in both training and test sets, leading to overly optimistic performance estimates. Therefore, the dataset was ordered by cycle index. Specifically, the data from the first 36 cycles (approximately 80 % of the total data) were utilized for model

training, while the remaining cycles were reserved as an independent test set.

This study employed four algorithms (Fig. S1), included two ensemble ML models, as random forest (RF) and extreme gradient boost (XGBoost), and two DL models, as multiple layer (MLP) and long-short term memory (LSTM). RF is one of the bagging methods, which combines predictions from multiple decision trees to reduce overfitting and improve accuracy (Li et al., 2018). XGBoost, a type of gradient boosting machines (GBM), use a boosting strategy that builds models sequentially with each iteration correcting the errors of previous one to enhance overall performance (Chen & Guestrin et al., 2016). The MLP relies on neural networks composed of interconnected nodes and layers to estimate output values. LSTM, a specialized form of recurrent neural network (RNN), is designed to model sequential and time-dependent data effectively through employing unique gating mechanisms that selectively retain, forget, or output information (Niu et al., 2022).

The ML/DL models developed in this study are based on the open-source programming language Python. The versions of the representative packages are Python = 3.9.16, NumPy = 1.4.4, Pandas = 2.2.2. RF, XGoost, and MLP model were conducted with Python via the Scikit-learn package (Scikit-learn = 1.2.1), and LSTM mode was performed by TensorFlow = 2.10.1, and Keras = 2.10.0. Further details of the model environmental setting are provided in Table S3. Additionally, the configuration of LSTM model was also listed in Table S4.

After establishing the initial ML/DL models, hyperparameter tuning was conducted to reduce the risk of overfitting and balance predictive performance and model stability by minimizing the loss function. Overfitting refers to a model's tendency to learn noise or anomalies in the training data, thereby reducing its ability to generalize to unseen data (Dietterich et al., 1995). In this study, a restricted grid search approach, one of the most widely used methods for hyperparameter optimal optimization in ML/DL models, was employed to reduce the result variance and identify the optimal hyperparameters combination (Menapace et al., 2021). In addition, an early stopping strategy was applied to prevent overfitting and over-training, Within the hyperparameter turning, it set, 36 cycles were used for training and 6 cycles for validation, and the best configuration was selected by minimizing the validation RMSE on the cycle-based validation set. After tuning, the final model was refit on the combined training and validation data and evaluated once on the unseen test set. The detail search spaces used in restricted grid search for each model are provided in Table S5.

2.2.3. Model evaluation and interpretation

Spiess et al. (2010) noted that the coefficient of determination (R^2) alone is an inadequate measure for assessing the goodness of fit in nonlinear models. A robust model evaluation should incorporate multiple metrices to comprehensively assess the predictive performance. Therefore, in this study, the predictive performance of all ML/DL models was validated using R^2 root mean squared error (RMSE), and mean absolute error (MAE), which are among the most commonly adopted metrics for regression tasks (Zhu et al., 2023), as defined in the following equations:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{\sum_{i=1}^{n}(\hat{y}_i - \bar{y}_i)^2} \quad (5)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{n}} \quad (6)$$

$$MAE = \frac{\sum_{i=1}^{n}|\hat{y}_i - y_i|}{n} \quad (7)$$

where y_i represents the observed values from the experiment, $\bar{y}$ represents the mean values of the experiment, the$\hat{y}$represents the predicted values, and the n represents the total number of paired observed values and predicted values.

To further verify model stability, a cycle-level 5-fold cross-validation (CV) was conducted using sequential splits, reporting the mean ± standard deviation of R^2, RMSE, and MAE. In each fold, models were trained on earlier cycles and evaluated on subsequent cycles. This approach strictly preserves the temporal structure of the data and prevents look-ahead bias (data leakage) across folds.

For model interpretation, Shapley Additive Explanations (SHAP) was employed to quantify the contributions of input features to the model outputs and to decode the underlying physicochemical logic of the black-box models. Based on game theory, SHAP provides a unified measure of feature importance by evaluating the marginal contribution of each feature across all possible feature combinations, (Yun et al., 2025; Chen et al., 2024). This approach enables a consistent and interpretable understanding of how individual input features influence the model's predictions.

3. Results and discussion

3.1. Data Preprocessing and Feature selection

Before conducting feature selection, the reliability of the imputed data was verified. As described in Section 2.2.1, a sensitivity analysis was performed by comparing the prediction performance using datasets processed with the proposed a first-order Lagrange interpolation (equivalent to linear interpolation) and a higher-order Lagrange- polynomial interpolation (degree ≤ 3).

The high-order polynomials introduced non-physical overshooting during the steep initial conductivity drop, even generating impossible negative conductivity values (e.g., dropping below -392 μS/cm) (Fig. S2 (b)). These artifacts injected noise into the training data, misleading the learning models. In contrast, first-order polynomials presented relative stable performance and didn't produce negative conductivity values throughout the cycle. Furthermore, the predictive results reveal an obvious divergence in model stability (Table S6). The first-order interpolation consistently supported better generalization performance across all models. Taking RF model as an example, the model achieved a test R^2 of 0.81 with the first-order method, whereas the performance severely degraded to an R^2 of 0.37 when using high-order imputation. Besides, given the low missing-data ratio (<4.7 %) and short gap duration, the first-order approach provided a physically consistent reconstruction without introducing artificial variance. Therefore, first-order Lagrange (linear) interpolation was confirmed as the robust choice for subsequent feature analysis and model development.

In this study, the initial selection of input features was based on MCDI theoretical mechanisms (Tang et al., 2017; Wang et al., 2019) and prior experimental experience (Hou et al., 2013; Liu et al., 2019; Chen et al., 2025). The selected input features included time (T), influent conductivity (EC_{in}), pH, and current (A). PCC analysis was performed to quantify the relationship between each input feature and the target output variable, i.e., effluent conductivity as a function of time (EC_{out}). The correlation coefficient (r), weas calculated using the Python pandas library and are presented in Table S7.

Among the selected features, current exhibited the strongest positive correlation with EC_{out}, with a r of 0.85 (Fig. 3). As seen, a constant voltage was applied during the charging periods. According to Ohm's law for charge transport in an electric field, the resistance to charge transport increases as ion concentration decreases, which in turn reduces the current (Biesheuvel et al., 2009). In addition, pH also showed a strong positive correlation with EC_{out} (r=0.68), which aligns with the theoretical model proposed by Dykstra et al. (2017), who predicted the temporal distribution of pH, demonstrating an increase during the adsorption phase and a decrease during desorption. Similarly, Yu et al. (2018) reported that the accumulation of H^+ from the time of peak pH to the end of the charging period was caused by oxidation of carbon electrodes, further contributing to the observed pH swing. These

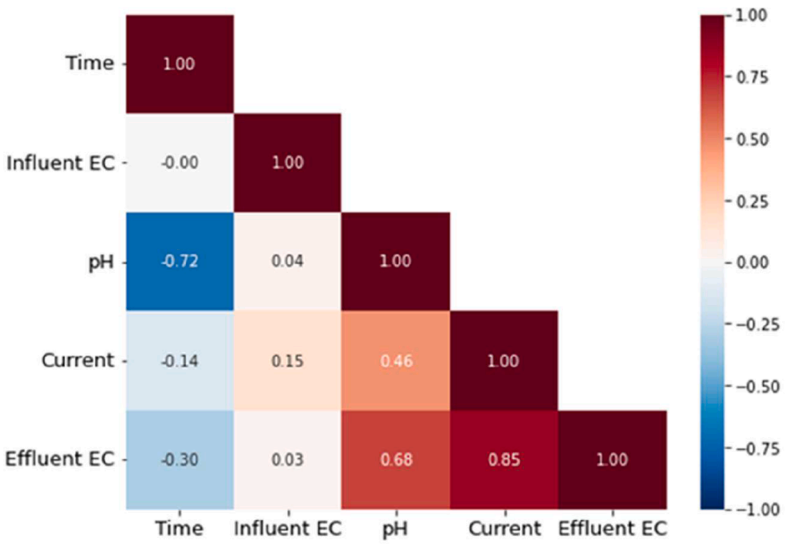


Fig. 3. Pearson correlation matrix showing the relationships among candidate input features and the target output (Effluent EC). This heatmap visualizes the pairwise Pearson correlation coefficients among Time, Influent EC, pH, Current, and Effluent EC. The color gradient from blue to red indicates negative to positive correlations, respectively.

findings support the positive correlation between pH and EC_{out}. In addition, EC_{out} was expected to decrease over time due to ion adsorption during the charging period. However, the observed correlation coefficient between T and EC_{out} was only –0.30. This relatively weak negative association could be attributed to the nonlinear nature of the conductivity decline, and the concurrent influence of other operational variables such as current and pH. Additionally, the use of fixed-interval sampling (every 3 seconds) may have limited the model's ability to capture rapid variations, resulting in an attenuated correlation signal.

It is worth noting that EC_{in} exhibited a negligible linear correlation with EC_{out} (r=0.03). This statistical observation does not imply that EC_{in} is physically unimportant. Rather, it reflects the nature of the batch process where EC_{in} remains relatively constant while EC_{out} varies dynamically over time. In this study, the model was designed to predict the cycle-level, time-dependent EC_{out} profile within a single charging cycle. Consequently, the variation of EC_{out} was primarily governed by operational conditions (e.g., current and charging time), whereas the effect of the EC_{in} was comparatively weak in this time-dependent context. To mitigate feature redundancy and overfitting risks in the ML model, candidate input variables with r above 0.1 were retained, as correlations above this threshold are generally regarded as indicative of statistically meaningful correlations (Schober et al., 2018). Based on this criterion, Time, Current, and pH were selected as model inputs.

To further validate the exclusion of EC_{in} is reasonable, a comparative analysis was performed with and without EC_{in} (Fig. S3 and Fig. S4). While including EC_{in} yielded marginal improvements for ensemble ML models (e.g., RF and XGBoost), it notably degraded the performance of MLP and LSTM models. This inconsistency suggests that for DL models, EC_{in} may introduce noise or complexity that outweighs its predictive value under the current dataset constraints. To ensure a robust and parsimonious feature, the decision to exclude EC_{in} was upheld. This approach avoids feature redundancy without substantially compromising the overall predictive skill. Furthermore, three input features were collected and used yielding an SFR greater than 1817, which exceeds the recommended threshold of 100 (Zhu et al. 2023) to ensure an adequate SFR and minimize overfitting risk.

3.2. Prediction model comparisons

After data preprocessing and Feature selection, the dataset was split into training and testing sets using a cycle-based chronological strategy to avoid information leakage. EC_{out} was selected as the target variable and was predicted using four different algorithms, including RF, XGBoost, MLP, and LSTM. Detail information on the original hyperparameters setting and predictive results for each model was provided in Table S8.

In Fig. 4a-d, the x-axis represented the actual value and the y-axis denotes the predicted value. Good predictive performance is indicated by minimal deviation from the 1:1 reference line, along with high R^2 and low MAE and RMSE values. Among these models, the RF model achieved a test R^2 of 0.81, while the XGBoost model achieved a test R^2 of 0.82. Both models exhibited minimal deviation from the reference line, indicating their capability to capture the EC_{out} in MCDI process dynamics effectively (Table 1).

Conversely, the DL models exhibited poor performance. The MLP model produced a test R^2 of 0.50 (Table 1). The LSTM model, despite reaching an R^2 of 0.71, showed signs of significant overfitting, as evidenced by the large discrepancy between training RMSE (15.01 μS/cm) and testing RMSE (42.46 μS/cm). Most data points for these DL models fell noticeably below the 1:1 reference line, suggesting insufficient model fitting (Wang et al., 2025) This limitation likely stems from the relatively small dataset size, which may not provide sufficient temporal dependency to train complex deep sequence architectures effectively without extensive regularization.

To further verify model stability, a cycle-level 5-fold cross-validation was conducted using sequential splits, with the mean ± standard deviation of R^2, RMSE, and MAE (Table S9). The results reveal a stark contrast in robustness. RF and XGBoost demonstrated remarkable stability, yielding the highest mean R^2 of 0.88 (± 0.05). In contrast, the LSTM model failed to generalize consistently, yielding a negative mean R^2 of -0.084 with a relatively high standard deviation (0.50). This instability confirms that the current feature set and data scale are better suited for ensemble ML models than for DL models. Overall, the results obtained with the original hyperparameters setting indicated the ensemble ML models provided better fits and stronger predictive performance than the DL models, as reflected by the higher R^2 values and lower RMSE and MAE values in testing phase (Fig. 4e and f). Notably, the performance gap between training and testing, as reflected by differences in R^2, RMSE, and MAE, revealed the model generalization under the default settings. Therefore, hyperparameter tuning was subsequently performed to improve predictive accuracy and enhance model generalization.

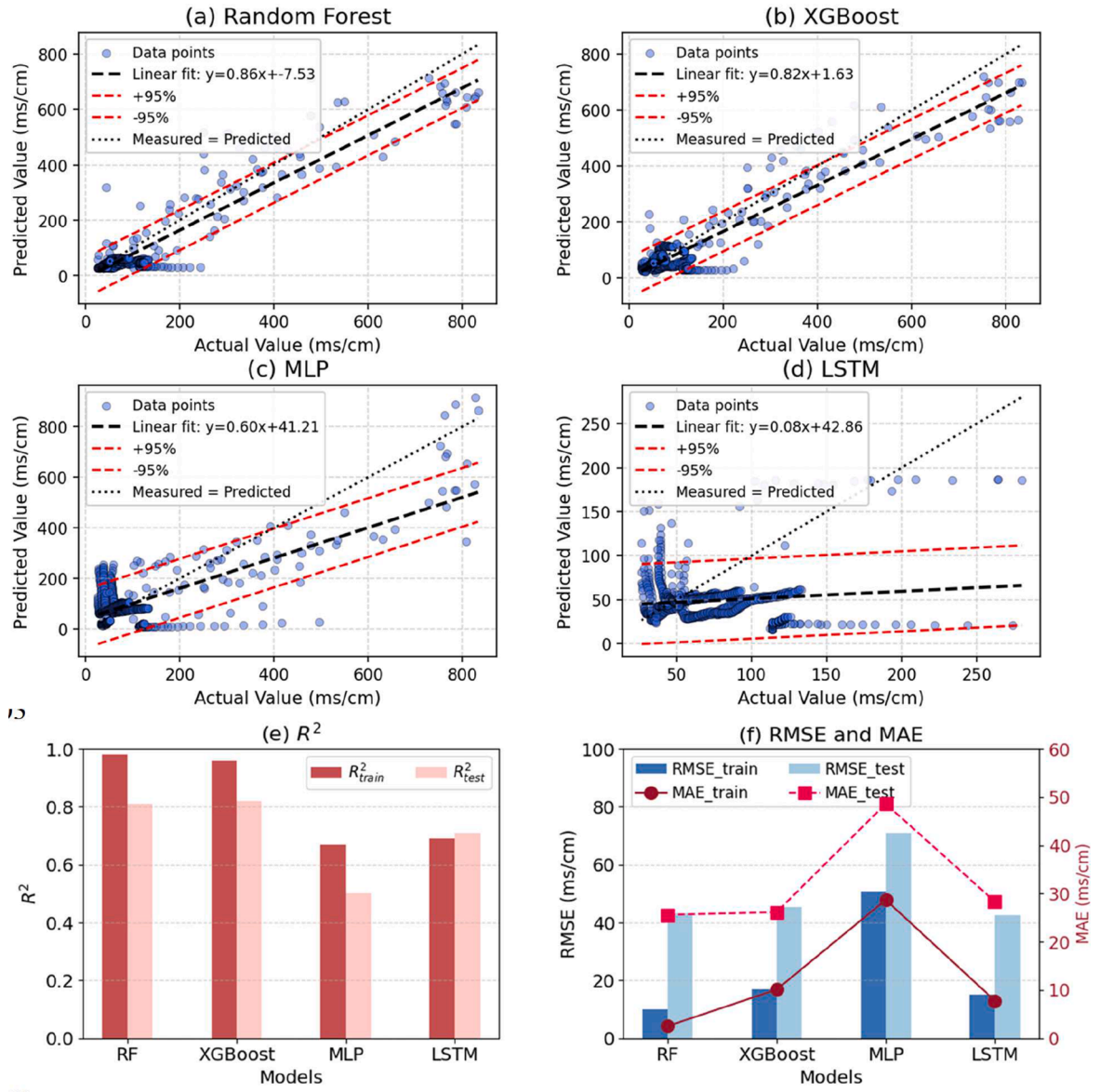


Fig. 4. Model performances of four different ML/DL models for prediction EC_{out} under original hyperparameter setting. (a) RF, (b) XGBoost, (c) MLP, (d) LSTM, and (e, f) the comparison of the R^2, the RMSE and MAE performance on both training and testing datasets.

Table 1
Comparison of model performances in terms of R2, RMSE and MAE values under original hyperparameter.

model	R^2		RMSE		MAE	
	train	test	train	test	train	test
RF	0.98	0.81	10.10	42.32	2.50	25.60
XGBoost	0.96	0.82	16.90	42.06	10.72	26.16
MLP	0.67	0.50	50.78	73.55	28.71	48.63
LSTM	0.69	0.71	15.01	42.46	7.76	28.32

Table 2
Comparison of model performances in terms of R^2, RMSE and MAE values under hyperparameter tuning.

model	R^2		RMSE		MAE	
	train	test	train	test	train	test
RF	0.95	0.79	17.22	36.25	9.91	22.92
XGBoost	0.94	0.78	17.45	37.34	11.14	23.20
MLP	0.92	0.77	20.54	38.00	13.45	25.62
LSTM	0.54	0.49	13.80	36.44	8.8	22.82

3.3. Model optimization under hyperparameter tuning

Adjusting the hyperparameters of each algorithm was essential for balancing both predictive accuracy and generalization ability. A restricted gird search approach was employed to systematically explore different combinations of hyperparameters and to identify the optimal settings based on model evaluation metrics. Table 2 lists the prediction performance of each algorithm, and other the best hyperparameter configurations was summarized in Table S10.

Among these four algorithms, RF and XGBoost consistently maintained their statues as the top-performing models. For RF model, although the test R^2 value showed a marginal decrease from 0.81 to 0.79, the model accuracy improved significantly, as indicated by the RMSE and MAE. The RF model achieved the lowest error metrics, with RMSE decreasing from 42.32 to 36.25 μS/cm and MAE from 25.60 to 22.92 μS/cm. This phenomenon indicates that the tuned models

demonstrate a better balance between bias and variance, offering superior robustness for practical concentration predictions despite the slight trade-off in the correlation coefficient. Moreover, the reduced performance gap between the training and testing phases (Fig. 5e and f) confirms that hyperparameter tuning enhanced the models' generalization ability. These findings aligned with the observations of Kernbach et al. (2021), who reported that the tuned models are less prone to overfitting and exhibit better generalization.

Notably, Hyperparameter tuning yielded contrasting outcomes for the DL models (Fig. 5c, 5d, and Table S9). The MLP model exhibited substantial improvement. Its training R^2 increased markedly from 0.67 to 0.92, and the test R^2 surged from 0.50 to 0.77. The training RMSE plummeted from 50.78 to 20.54 μS/cm, bringing predicative performance much closer to that of the ensemble models. This demonstrates that with proper tuning, MLP can capture the process dynamics effectively. Conversely, the LSTM model failed to benefit from the optimization within the search space. Its performance actually degraded, with the training R^2 falling to 0.54 and test R^2 dropping to 0.49. Furthermore, the observed divergence between the relatively low R^2 and the competitive RMSE and MAE reflects the LSTM model's limitation on this specific dataset. Due to the limited data scale or sensitivity to hyperparameter initialization, the LSTM model struggled to capture the sharp dynamic trends of the initial adsorption peak. Instead, the model tended toward mean-reverting predictions. The MCDI experiments in this study operates under constant voltage mode, the ion transition starts with a rapid adsorption phase and gradually moves toward a saturation phase, where the concentration gradient diminishes, and the EC_{out} becomes relatively stable. While these mean-reverting predictions align with the stable physical behavior of EC_{out} during the later stages of the cycle, leading to competitive RMSE and MAE values, the results in a low R^2 because the model fails to explain the critical dynamic transitions of the charging process.

Although MLP showed great potential after tuning, the ensemble ML models (RF and XGBoost) remained the most robust candidates overall (Fig. 6). While MLP presents R^2 of 0.77, which narrowed the performance gap, the ensemble models consistently yielded stable and higher test R^2 values (0.79 and 0.78), lowest RMSE and MAE, and demonstrated the lowest variance in the earlier cross-validation analysis. This suggests

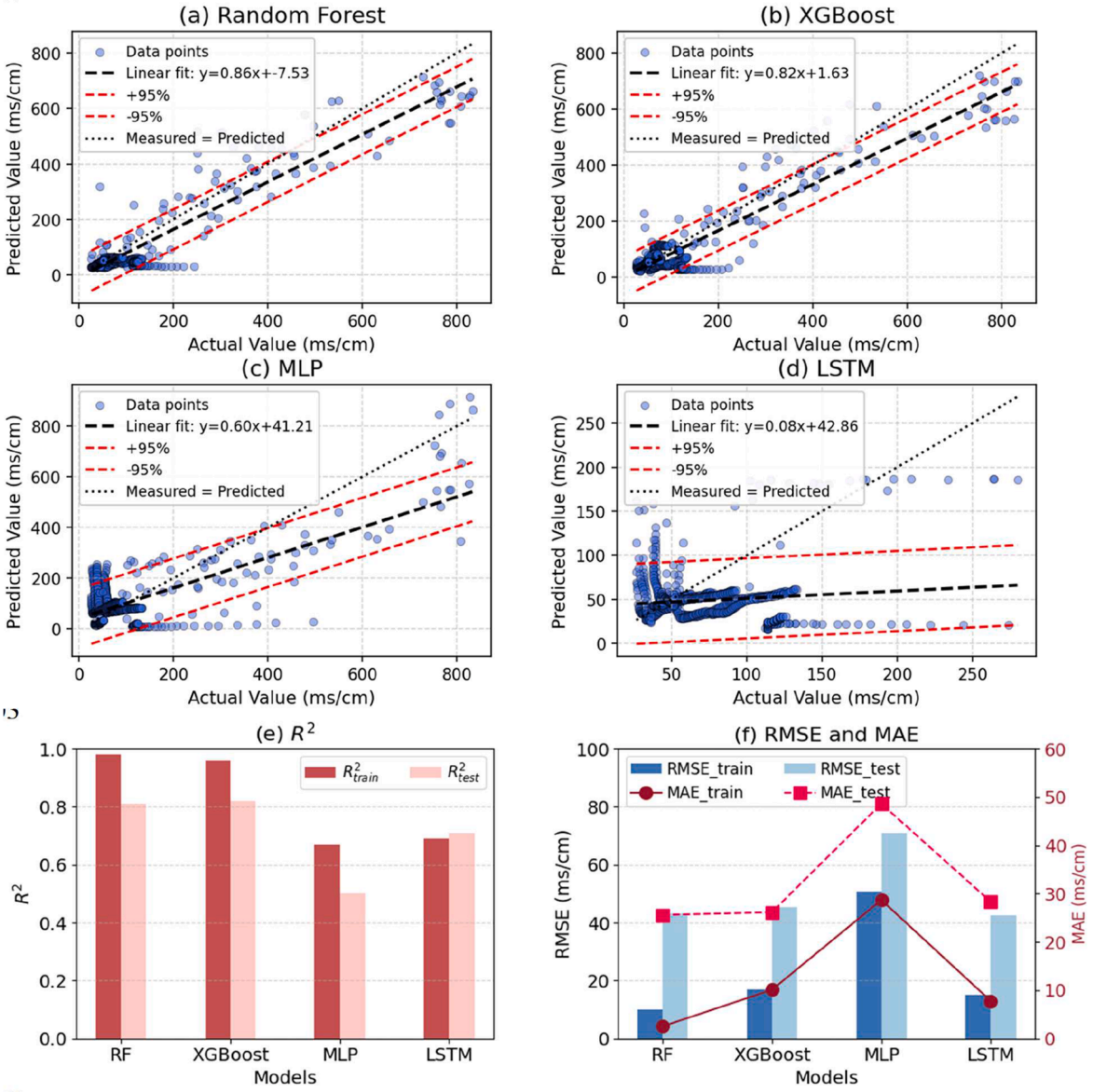


Fig. 5. Model performances of four different ML/DL models for prediction EC_{out} under hyperparameter tuning. (a) RF, (b) XGBoost, (c) MLP, (d) LSTM, and (e, f) the comparison of the R^2, the RMSE and MAE performance on both training and testing datasets.

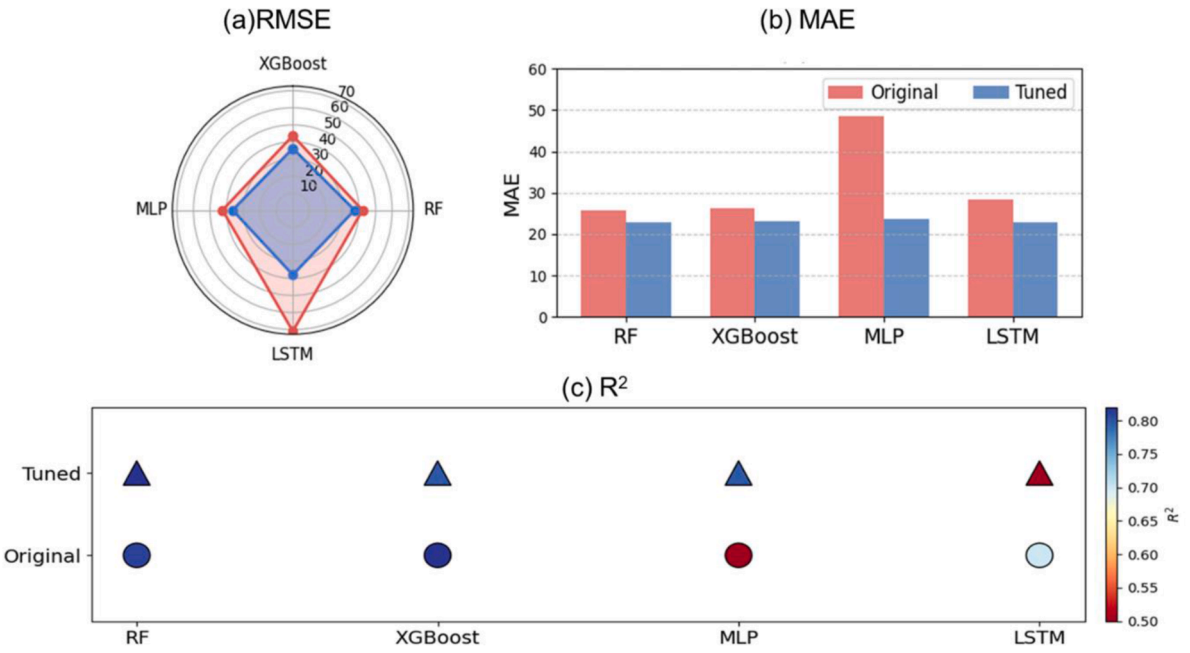


Fig. 6. Performance comparison of different ML/DL models using three metrics, R^2, RMSE, and MAE on training datasets. (a) RMSE. The blue point represented the hyperparameter under original setting, and the red point represented the results on optimal hyperparameter. (b) MAE. The red bar graph showed the performance before setting hyperparameter tuning, and the blue bar graph showed performance after using hyperparameter tuning. (c) R^2. Triangles represent the result after hyperparameter tuning and circles represent the result before hyperparameter tuning. The size of the shape and the shade of the color reflect the relative magnitude of the values.

that for experimental data with limited sample size, ensemble ML methods, which aggregate predictions from multiple weak learners, offer a more reliable generalization capability than deep neural networks, which are more data-hungry and prone to local optima.

Furthermore, the discrepancy in model performance can be attributed to the intrinsic nature of the dataset and the physical characteristics of the MCDI process. First, regarding feature types, the dataset comprises tabular data with heterogeneous variables (e.g., current, pH, and time) that possess distinct physical units and statistical distributions. Tree-based ML models naturally handle such irregular decision boundaries in tabular features, whereas deep neural networks often struggle to learn effective representations from tabular data without extensive preprocessing or specialized embeddings (Grinsztajn et al., 2022). Second, concerning temporal dependency, the MCDI operation is a batch process where the system state effectively resets after regeneration. The effluent conductivity is predominantly governed by the instantaneous electrochemical state rather than long-term historical dependencies. Consequently, the complex recurrent architecture of LSTM, which designed to capture long-range sequence patterns, is theoretically unsuitable for this task. Instead of learning meaningful temporal dynamics, the LSTM model likely overfitted to the short-term sensor noise, as evidenced by its instability in the cross-validation analysis.

XGBoost relies on a sequential residual learning strategy, in which each tree is trained to correct the errors of the previous one. While this approach effectively reduces bias, it also makes the model more susceptible to overfitting, particularly in the presence of noisy training data. In contrast, RF constructs multiple decision trees in parallel using randomly sampled subsets of data and features. Because each tree is trained independently and the final output is obtained by averaging predictions, RF model is inherently more robust to overfitting and less sensitive to the variance of individual trees (Segal et al., 2004, Chang et al., 2022).

3.4. Performance validation and interpretation of the RF model

In the previous subchapter, the optimized RF presented the highest R^2 and the lowest RMSE and MAE values among all selected ML/DL models. Based on its predictive performance and model characteristic, RF was chosen as the preferred model for further in-depth analysis. The regression results and time-series comparisons showed that the RF model exhibited a strong predictive trend with most data points falling within the ±95 % confidence interval. The residuals were closely distributed along the diagonal, indicating good normality. The fitted line ($y = 0.86x + -4.99$) was closely aligned with the 1:1 reference line, suggesting that the model captured the relationship between observed and predicted values accurately with minimal bias. In addition, the temporal profile of predicted versus actual EC_{out} (Fig. S5) demonstrated good agreement across the entire time series.

Since ML models are often regarded as “black boxes” with limited interpretability, SHAP analysis was employed to clarify the contribution of individual features (Kong et al., 2025). The bar plot in Fig. 7a shows the mean absolute SHAP values, representing the average impact of each input variable on the model’s predictions. Among the selected inputs, Time emerged as the most dominant feature (SHAP value = 26), followed by Current (8.5) and pH (7.2). This ranking aligns with the known behavior of MCDI systems, where EC_{out} dynamically governed by the interplay between adsorption performance and electro-sorption capacity. To gain deeper insights, a SHAP summary plot was generated (Fig. 7b) to visualize how individual feature values affect the predicted EC_{out}, revealing both the direction and magnitude of each contribution. In this plot, each point represents a single instance. Color indicates the relative magnitude of the feature value (red = high, blue = low), while the position on the x-axis shows the SHAP value, indicating whether the feature drives the prediction higher or lower.

Specifically, higher current values tended to show positive SHAP contributions. This aligns with the Nernst-Planck theory, where current serves as the direct driving force for electromigration. Although the system operated under constant voltage, causing the current to decline naturally as resistance increased due to ion be removed and adsorbed, the model correctly identified that higher instantaneous current correlates with a stronger ion flux, validating its ability to capture the fundamental driving force of the separation process.

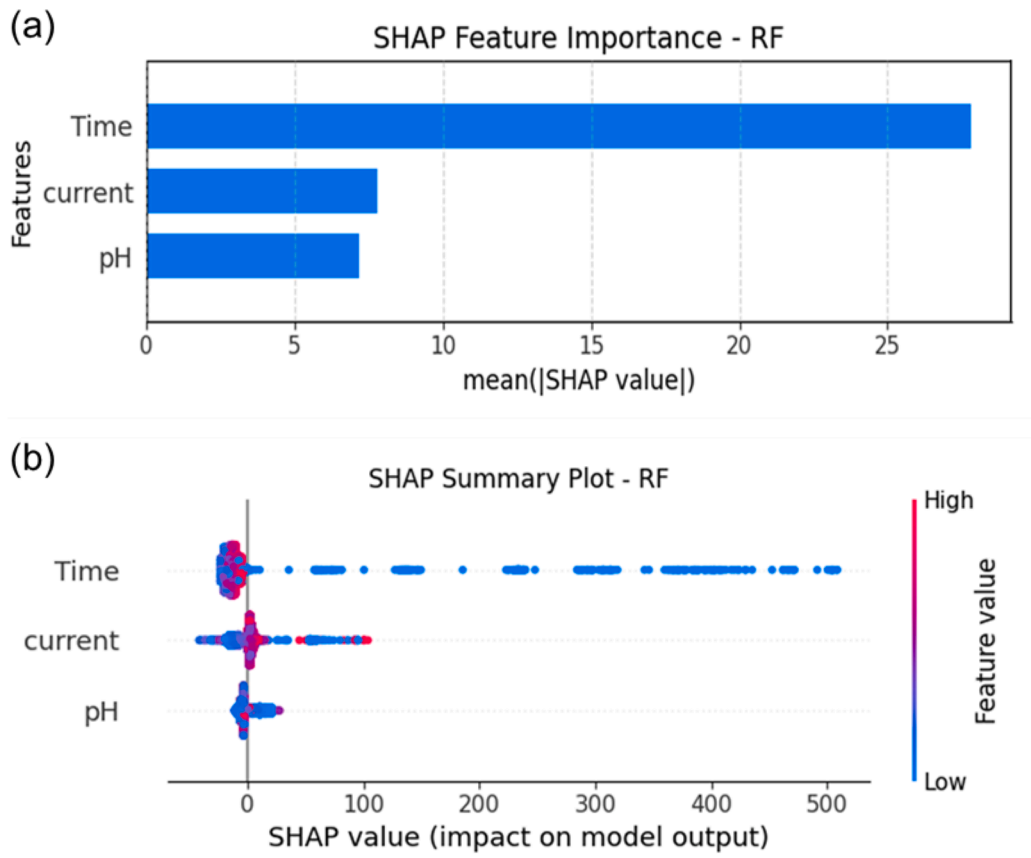


Fig. 7. SHAP-based contribution analysis of input features to the predicted effluent conductivity using the RF model. (a) SHAP feature importance bar chart showing the average magnitude of each feature's contribution to the model output. (b) SHAP summary plot illustrating both the magnitude and direction of feature contributions. Each dot represents a single prediction, and the color gradient reflects the original feature value, with red indicating higher values and blue indicating lower values.

Notably, higher Time values (red points) were consistently associated with negative SHAP values, driving the predicted EC_{out} lower. Physically, this indicates that the model utilizes "Time" not merely as a counter, but as a proxy for the state of sorption saturation. The high SHAP impact reflects the transition from the initial rapid adsorption phase (kinetically controlled) to the later saturation phase, where the concentration gradient diminishes. Last but not the least, pH exhibited a significant contribution (SHAP value = 7.2), comparable to that of Current (8.5). This is a critical finding because pH is often treated as a secondary or constant parameter in theoretical models (Liu et al., 2022, Zhao et al., 2013).

Its significance in the SHAP analysis implies that the model is capturing complex physicochemical interactions beyond simple electrosorption. Specifically, since the ionization of functional groups on carbon electrodes and the occurrence of faradaic side reactions are heavily pH-dependent, the model's reliance on pH suggests it is strictly accounting for these surface charge fluctuations and electrochemical stability factors to maintain prediction accuracy. By assigning substantial weight to pH, the RF model demonstrates an ability to correct for non-ideal chemical conditions that purely kinetic models might overlook.

4. Conclusion

This study established a comprehensive ML model framework for predicting the performance of a scaled-up MCDI system, encompassing data preprocessing, model development, performance evaluation, and model interpretation. Data preprocessing served as a crucial first step to ensure model reliability, including outlier removal, gap filling, and data normalization to preserve the integrity of the training process. In model development, hyperparameter optimization was conducted using a restricted grid search approach to systematically tune each model and identify optimal configurations based on performance metrics (R^2, RMSE, and MAE), while minimizing overfitting. It was found that among the four models, the RF model exhibited the best predictive performance (EC_{out}), with an R^2 of 0.79, RMSE of 36.25 μs/cm, and MAE of 22.92 μs/cm in the test set. Model interpretability was enhanced through SHAP analysis, which quantified the influence of each input feature on the predicted EC_{out}. The results indicated that time was the most influential variable, followed by current and pH. This ranking was consistent with the experimental observations and known system behavior. Overall, this study provides the first demonstration of the strong potential of ML in the field of scaled-up MCDI systems. Building on this charging-phase prediction framework, future work will extend the model from within-cycle dynamics to cycle-to-cycle behavior, enabling prediction of full-cycle outcomes and process-level indicators. In particular, we will integrate operational variables and cycle-level performance metrics (e.g., salt removal efficiency, capacity, water recovery, and energy consumption) to support optimization of charging/discharging duration and operating conditions for improved overall system performance.

CRediT authorship contribution statement

Huei-Cih Liu: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Chia-Hung Hou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2026.125839.

Data availability

Data will be made available on request.

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